Performance Test — Executive Summary

12m 4s	+907 MB (+33%)	4915 MB	21%	91	0
Test Duration	Memory Growth	Peak Memory	Avg CPU	Avg FPS	Console Errors

Health Check

- Memory: possible leak — 2730→3636 MB (+33%), peak 4915 MB
- FPS: smooth — avg 91, min 37
- CPU: healthy — avg 21%, peak 49%
- Console errors: none captured

Per-Phase Comparison

Segment	Mem avg	Mem peak	CPU avg	FPS avg	FPS min
Phase 1: Globe / Satellite	3173 MB	4608 MB	23%	87	37
Phase 2: Mercator / Dark	3442 MB	4884 MB	21%	95	54

Test Configuration

Run

App: https://drop-black-demo.weather.com/

Desk: None (baseline)

Phase duration: 5 min per phase

Timebar test: Off

Machine & display

OS: macOS 25.6.0 (arm64) · CPU: Apple M3 Pro (12 cores) · RAM: 36 GB · Host: TWC-M2HGDHWL4V

Monitor: 1718×960 CSS px @ 1x · 30-bit color · 121 Hz (max ~121 FPS)

Browser: Chromium 151.0.7922.34 · Test viewport: 1718×960

Phases

- Phase 1: Globe / Satellite
- Phase 4: Mercator / Dark

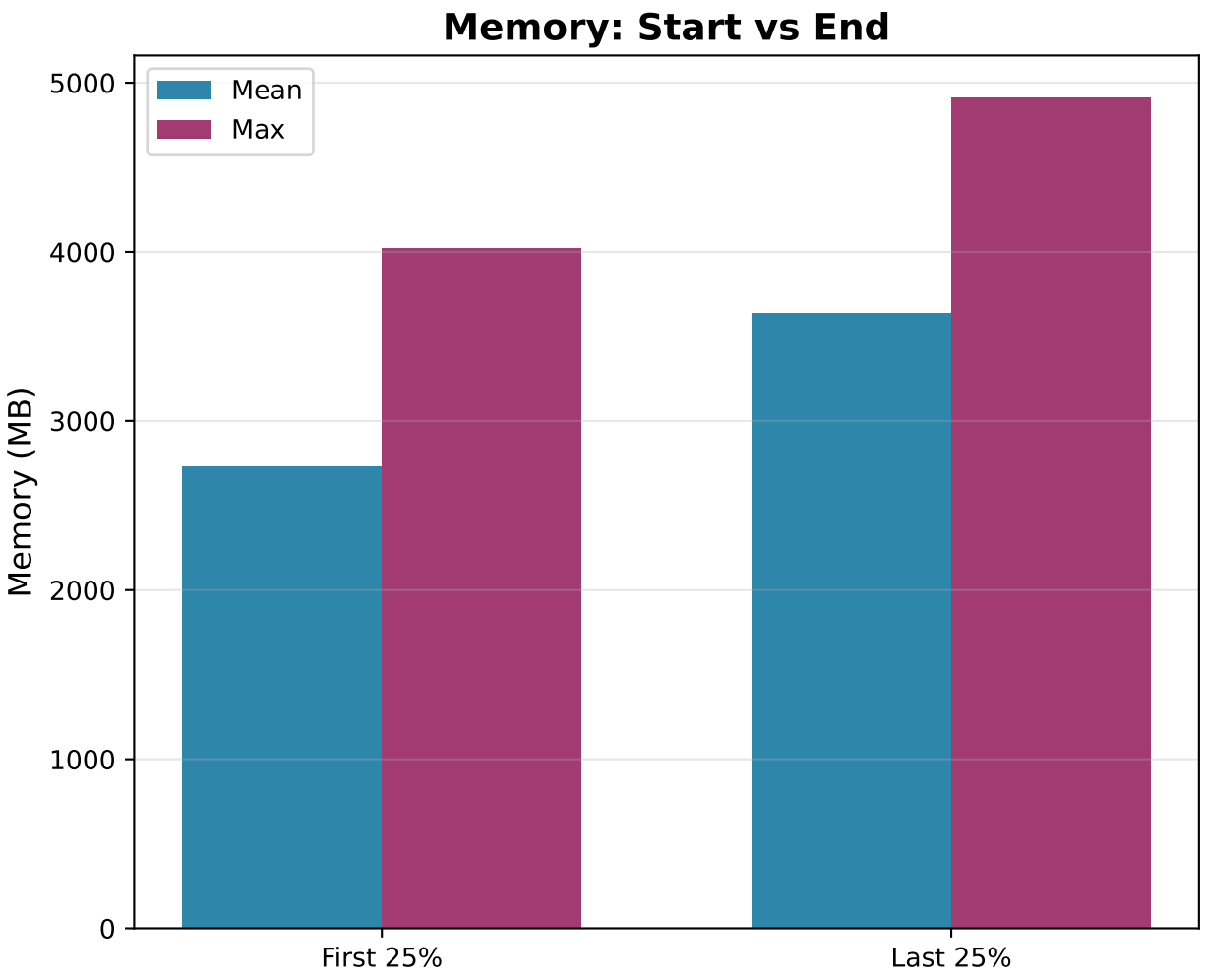
Layer (Layers tab)

Radar

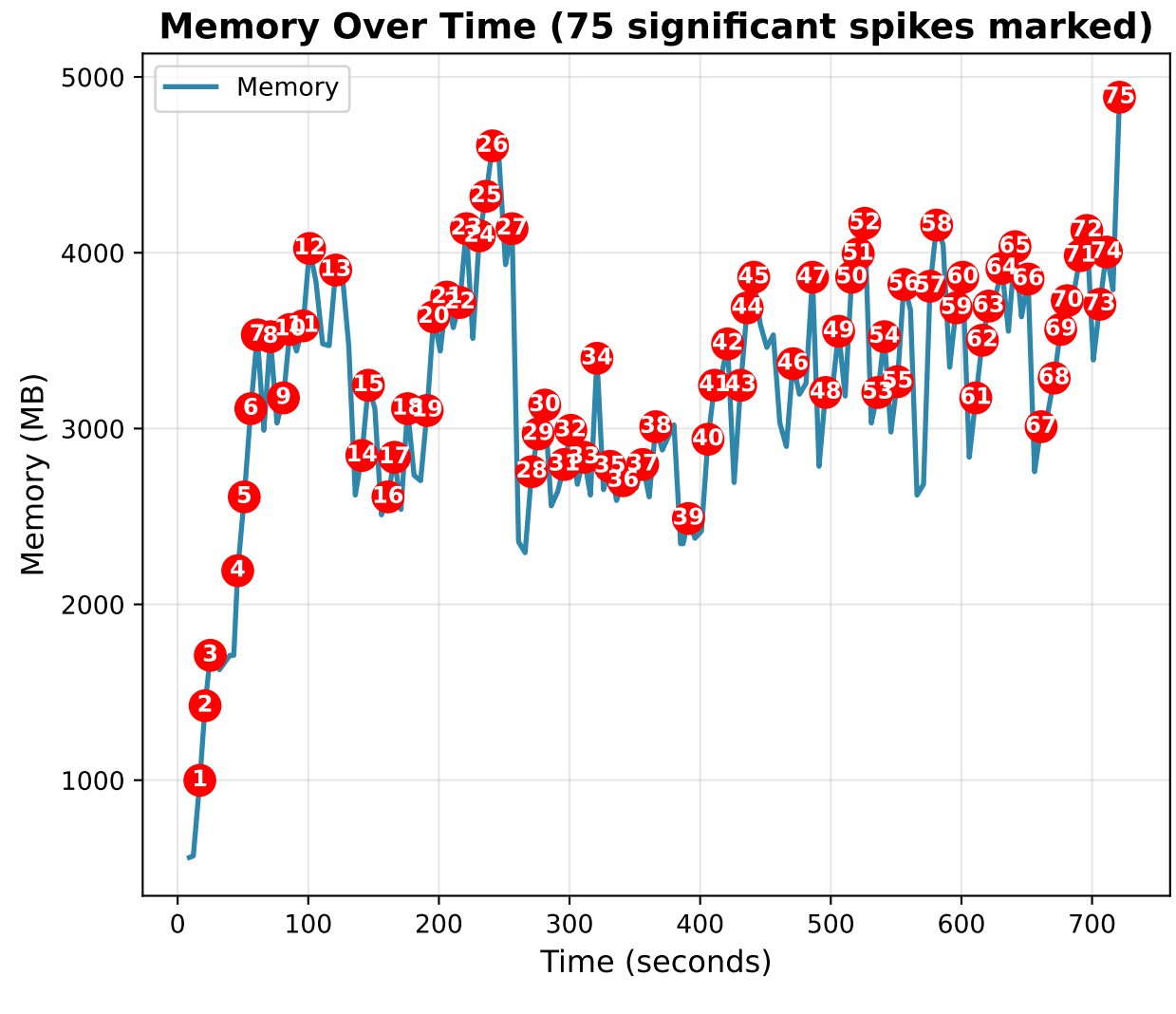
Overlays ON (4)

Lightning, Airports, Cman Buoys, Ship Reports

Memory Performance Analysis

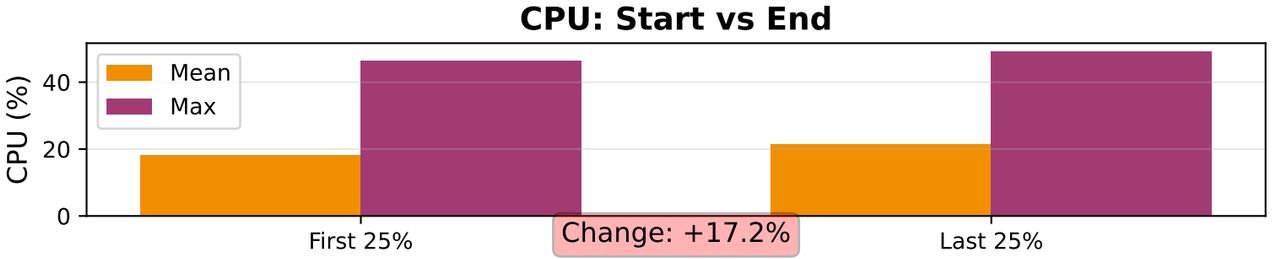


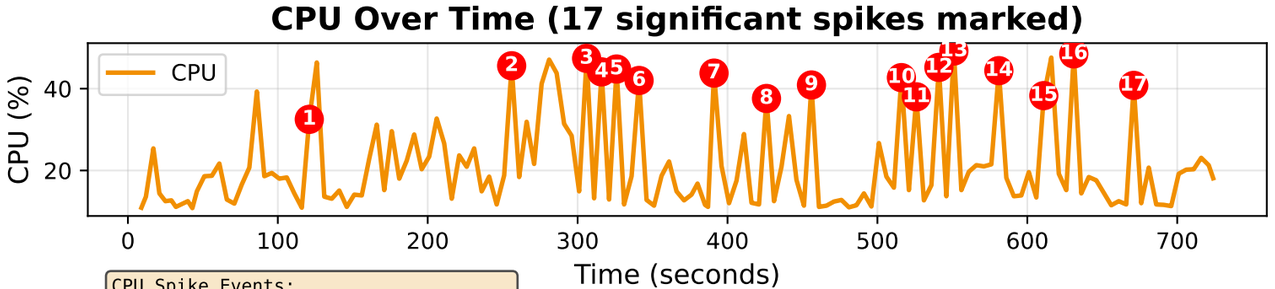
Change: +33.2%




```
Memory Spike Events:
  1. 17s: Radar enabled (999 MB)
  2. 21s: Lightning enabled (1423 MB)
  3. 25s: Airports enabled (1710 MB)
  4. 46s: Panning and zooming (2191 MB)
  5. 51s: Panning and zooming (2611 MB)
  6. 56s: Panning and zooming (3113 MB)
  7. 61s: Panning and zooming (3533 MB)
  8. 71s: Panning and zooming (3523 MB)
  9. 81s: Pan only (L/R) (3174 MB)
  10. 86s: Pan only (L/R) (3564 MB)
  11. 96s: Zoom only (in/out) (3584 MB)
  12. 101s: Zoom only (in/out) (4024 MB)
  13. 121s: Panning and zooming (3901 MB)
  14. 141s: Panning and zooming (2847 MB)
  15. 146s: Pan only (L/R) (3246 MB)
  16. 161s: Zoom only (in/out) (2611 MB)
  17. 166s: Zoom only (in/out) (2836 MB)
  18. 176s: Panning and zooming (3113 MB)
  19. 191s: Panning and zooming (3103 MB)
  20. 196s: Panning and zooming (3635 MB)
  21. 206s: Pan only (L/R) (3748 MB)
  22. 216s: Pan only (L/R) (3717 MB)
  23. 221s: Zoom only (in/out) (4137 MB)
  24. 231s: Zoom only (in/out) (4096 MB)
  25. 236s: Panning and zooming (4321 MB)
  26. 241s: Panning and zooming (4608 MB)
  27. 256s: Panning and zooming (4137 MB)
  28. 271s: Panning and zooming (2755 MB)
  29. 276s: Pan only (L/R) (2970 MB)
  30. 281s: Pan only (L/R) (3133 MB)
  31. 296s: Zoom only (in/out) (2796 MB)
  32. 301s: Zoom only (in/out) (2990 MB)
  33. 311s: Panning and zooming (2836 MB)
  34. 321s: Panning and zooming (3400 MB)
  35. 331s: Panning and zooming (2785 MB)
  36. 341s: Pan only (L/R) (2703 MB)
  37. 356s: Zoom only (in/out) (2796 MB)
  38. 366s: Zoom only (in/out) (3011 MB)
  39. 391s: Panning and zooming (2488 MB)
  40. 406s: Panning and zooming (2939 MB)
  41. 411s: Panning and zooming (3246 MB)
  42. 421s: Pan only (L/R) (3482 MB)
  43. 431s: Zoom only (in/out) (3246 MB)
  44. 436s: Zoom only (in/out) (3686 MB)
  45. 441s: Zoom only (in/out) (3860 MB)
  46. 471s: Panning and zooming (3369 MB)
  47. 486s: Pan only (L/R) (3860 MB)
  48. 496s: Pan only (L/R) (3205 MB)
  49. 506s: Zoom only (in/out) (3553 MB)
  50. 516s: Panning and zooming (3860 MB)
  51. 521s: Panning and zooming (3994 MB)
  52. 526s: Panning and zooming (4168 MB)
  53. 536s: Panning and zooming (3205 MB)
  54. 541s: Panning and zooming (3523 MB)
  55. 551s: Pan only (L/R) (3267 MB)
  56. 556s: Pan only (L/R) (3820 MB)
  57. 576s: Zoom only (in/out) (3809 MB)
  58. 581s: Panning and zooming (4157 MB)
  59. 596s: Panning and zooming (3686 MB)
  60. 601s: Panning and zooming (3860 MB)
  61. 611s: Panning and zooming (3174 MB)
  62. 616s: Panning and zooming (3502 MB)
  63. 621s: Pan only (L/R) (3697 MB)
  64. 631s: Pan only (L/R) (3912 MB)
  65. 641s: Zoom only (in/out) (4035 MB)
  66. 651s: Panning and zooming (3850 MB)
  67. 661s: Panning and zooming (3011 MB)
  68. 671s: Panning and zooming (3287 MB)
  69. 676s: Panning and zooming (3564 MB)
  70. 681s: Panning and zooming (3727 MB)
  71. 691s: Pan only (L/R) (3983 MB)
  72. 696s: Pan only (L/R) (4127 MB)
  73. 706s: Zoom only (in/out) (3707 MB)
  74. 711s: Zoom only (in/out) (4004 MB)
  75. 721s: Panning and zooming (4884 MB)
```

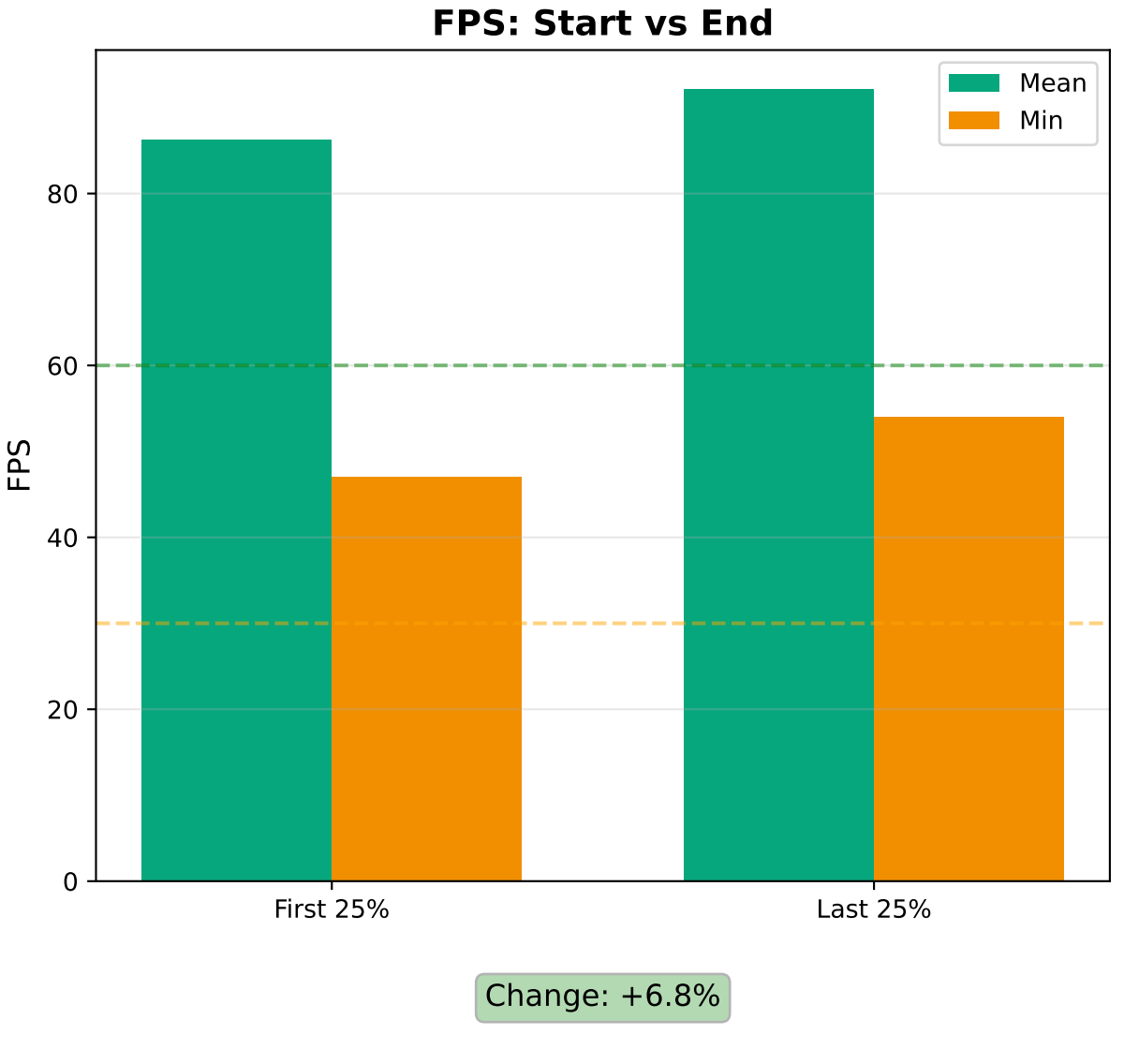
CPU Performance Analysis

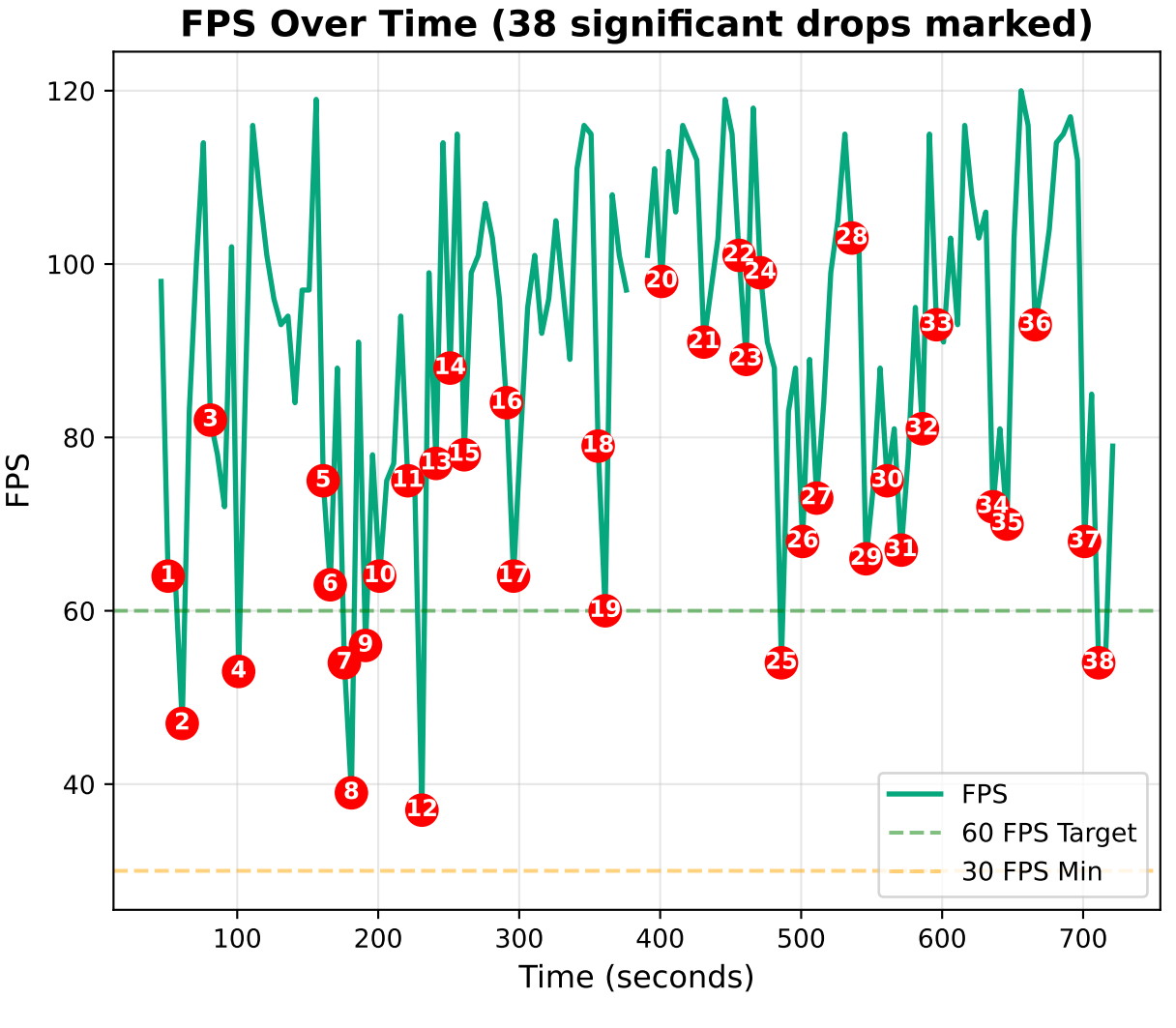





```
CPU Spike Events:
  1. 121s: Panning and zooming (32%)
  2. 256s: Panning and zooming (46%)
  3. 306s: Panning and zooming (47%)
  4. 316s: Panning and zooming (44%)
  5. 326s: Panning and zooming (45%)
  6. 341s: Pan only (L/R) (42%)
  7. 391s: Panning and zooming (44%)
  8. 426s: Pan only (L/R) (38%)
  9. 456s: Panning and zooming (41%)
  10. 516s: Panning and zooming (43%)
  11. 526s: Panning and zooming (38%)
  12. 541s: Panning and zooming (45%)
  13. 551s: Pan only (L/R) (49%)
  14. 581s: Panning and zooming (44%)
  15. 611s: Panning and zooming (38%)
  16. 631s: Pan only (L/R) (48%)
  17. 671s: Panning and zooming (41%)
```

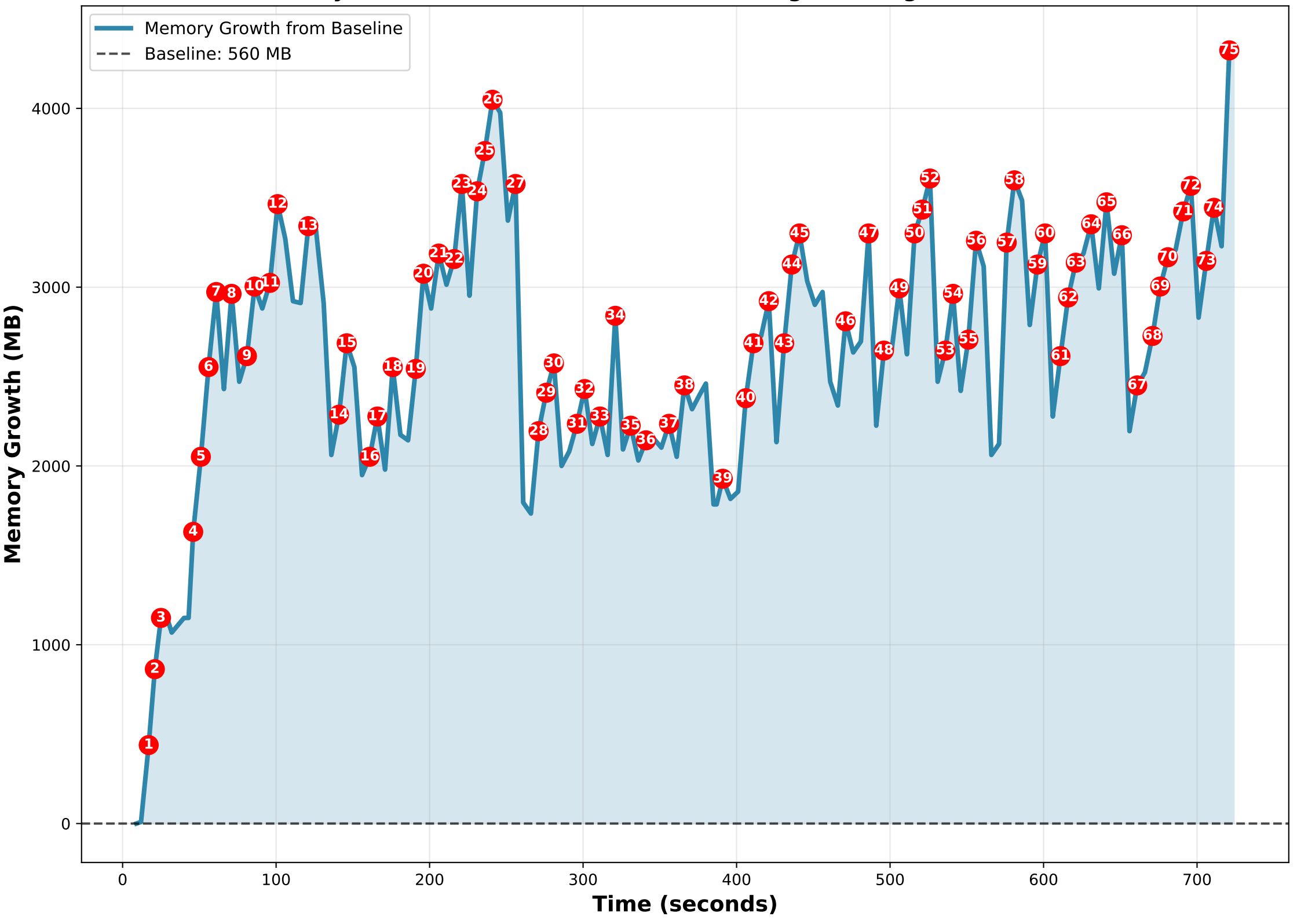
FPS Performance Analysis





FPS Drop Events:
1. 51s: Panning and zooming (64 FPS)
2. 61s: Panning and zooming (47 FPS)
3. 81s: Pan only (L/R) (82 FPS)
4. 101s: Zoom only (in/out) (53 FPS)
5. 161s: Zoom only (in/out) (75 FPS)
6. 166s: Zoom only (in/out) (63 FPS)
7. 176s: Panning and zooming (54 FPS)
8. 181s: Panning and zooming (39 FPS)
9. 191s: Panning and zooming (56 FPS)
10. 201s: Panning and zooming (64 FPS)
11. 221s: Zoom only (in/out) (75 FPS)
12. 231s: Zoom only (in/out) (37 FPS)
13. 241s: Panning and zooming (77 FPS)
14. 251s: Panning and zooming (88 FPS)
15. 261s: Panning and zooming (78 FPS)
16. 291s: Zoom only (in/out) (84 FPS)
17. 296s: Zoom only (in/out) (64 FPS)
18. 356s: Zoom only (in/out) (79 FPS)
19. 361s: Zoom only (in/out) (60 FPS)
20. 401s: Panning and zooming (98 FPS)
21. 431s: Zoom only (in/out) (91 FPS)
22. 456s: Panning and zooming (101 FPS)
23. 461s: Panning and zooming (89 FPS)
24. 471s: Panning and zooming (99 FPS)
25. 486s: Pan only (L/R) (54 FPS)
26. 501s: Zoom only (in/out) (68 FPS)
27. 511s: Zoom only (in/out) (73 FPS)
28. 536s: Panning and zooming (103 FPS)
29. 546s: Panning and zooming (66 FPS)
30. 561s: Pan only (L/R) (75 FPS)
31. 571s: Zoom only (in/out) (67 FPS)
32. 586s: Panning and zooming (81 FPS)
33. 596s: Panning and zooming (93 FPS)
34. 636s: Zoom only (in/out) (72 FPS)
35. 646s: Zoom only (in/out) (70 FPS)
36. 666s: Panning and zooming (93 FPS)
37. 701s: Zoom only (in/out) (68 FPS)
38. 711s: Zoom only (in/out) (54 FPS)

Cumulative Memory Growth Analysis

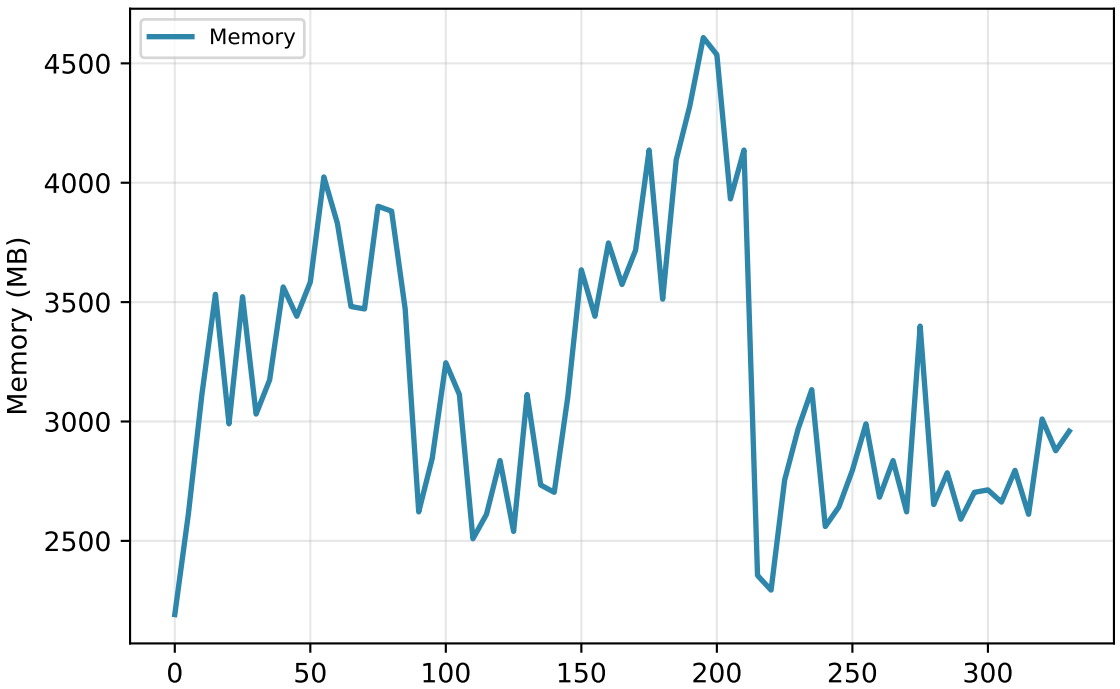


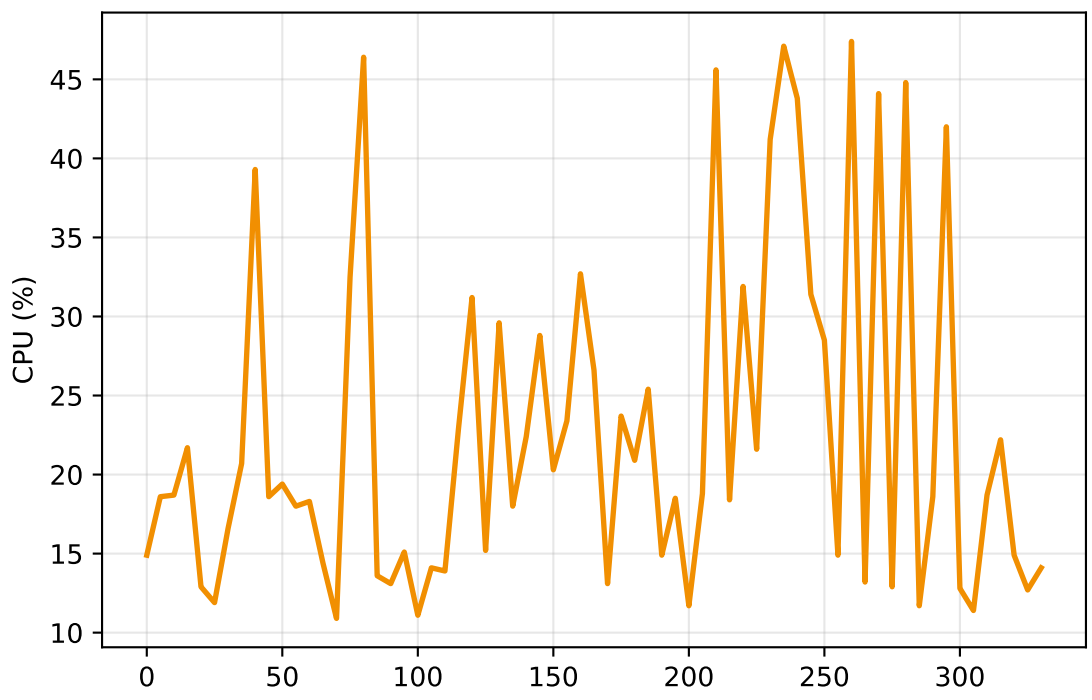

```
Significant Growth Events:
  1. 17s: Radar enabled (+430 MB)
  2. 21s: Lightning enabled (+424 MB)
  3. 25s: Airports enabled (+287 MB)
  4. 46s: Panning and zooming (+481 MB)
  5. 51s: Panning and zooming (+420 MB)
  6. 56s: Panning and zooming (+502 MB)
  7. 61s: Panning and zooming (+420 MB)
  8. 71s: Panning and zooming (+532 MB)
  9. 81s: Pan only (L/R) (+143 MB)
  10. 86s: Pan only (L/R) (+389 MB)
  11. 96s: Zoom only (in/out) (+143 MB)
  12. 101s: Zoom only (in/out) (+440 MB)
  13. 121s: Panning and zooming (+430 MB)
  14. 141s: Panning and zooming (+225 MB)
  15. 146s: Pan only (L/R) (+399 MB)
  16. 161s: Zoom only (in/out) (+102 MB)
  17. 166s: Zoom only (in/out) (+225 MB)
  18. 176s: Panning and zooming (+573 MB)
  19. 191s: Panning and zooming (+399 MB)
  20. 196s: Panning and zooming (+532 MB)
  21. 206s: Pan only (L/R) (+307 MB)
  22. 216s: Pan only (L/R) (+143 MB)
  23. 221s: Zoom only (in/out) (+420 MB)
  24. 231s: Zoom only (in/out) (+584 MB)
  25. 236s: Panning and zooming (+225 MB)
  26. 241s: Panning and zooming (+287 MB)
  27. 256s: Panning and zooming (+205 MB)
  28. 271s: Panning and zooming (+461 MB)
  29. 276s: Pan only (L/R) (+215 MB)
  30. 281s: Pan only (L/R) (+164 MB)
  31. 296s: Zoom only (in/out) (+154 MB)
  32. 301s: Zoom only (in/out) (+195 MB)
  33. 311s: Panning and zooming (+154 MB)
  34. 321s: Panning and zooming (+778 MB)
  35. 331s: Panning and zooming (+133 MB)
  36. 341s: Pan only (L/R) (+113 MB)
  37. 356s: Zoom only (in/out) (+133 MB)
  38. 366s: Zoom only (in/out) (+399 MB)
  39. 391s: Panning and zooming (+143 MB)
  40. 406s: Panning and zooming (+522 MB)
  41. 411s: Panning and zooming (+307 MB)
  42. 421s: Pan only (L/R) (+195 MB)
  43. 431s: Zoom only (in/out) (+553 MB)
  44. 436s: Zoom only (in/out) (+440 MB)
  45. 441s: Zoom only (in/out) (+174 MB)
  46. 471s: Panning and zooming (+471 MB)
  47. 486s: Pan only (L/R) (+604 MB)
  48. 496s: Pan only (L/R) (+420 MB)
  49. 506s: Zoom only (in/out) (+348 MB)
  50. 516s: Panning and zooming (+676 MB)
  51. 521s: Panning and zooming (+133 MB)
  52. 526s: Panning and zooming (+174 MB)
  53. 536s: Panning and zooming (+174 MB)
  54. 541s: Panning and zooming (+317 MB)
  55. 551s: Pan only (L/R) (+287 MB)
  56. 556s: Pan only (L/R) (+553 MB)
  57. 576s: Zoom only (in/out) (+1126 MB)
  58. 581s: Panning and zooming (+348 MB)
  59. 596s: Panning and zooming (+338 MB)
  60. 601s: Panning and zooming (+174 MB)
  61. 611s: Panning and zooming (+338 MB)
  62. 616s: Panning and zooming (+328 MB)
  63. 621s: Pan only (L/R) (+195 MB)
  64. 631s: Pan only (L/R) (+164 MB)
  65. 641s: Zoom only (in/out) (+481 MB)
  66. 651s: Panning and zooming (+215 MB)
  67. 661s: Panning and zooming (+256 MB)
  68. 671s: Panning and zooming (+205 MB)
  69. 676s: Panning and zooming (+276 MB)
  70. 681s: Panning and zooming (+164 MB)
  71. 691s: Pan only (L/R) (+215 MB)
  72. 696s: Pan only (L/R) (+143 MB)
  73. 706s: Zoom only (in/out) (+317 MB)
  74. 711s: Zoom only (in/out) (+297 MB)
  75. 721s: Panning and zooming (+1096 MB)
```

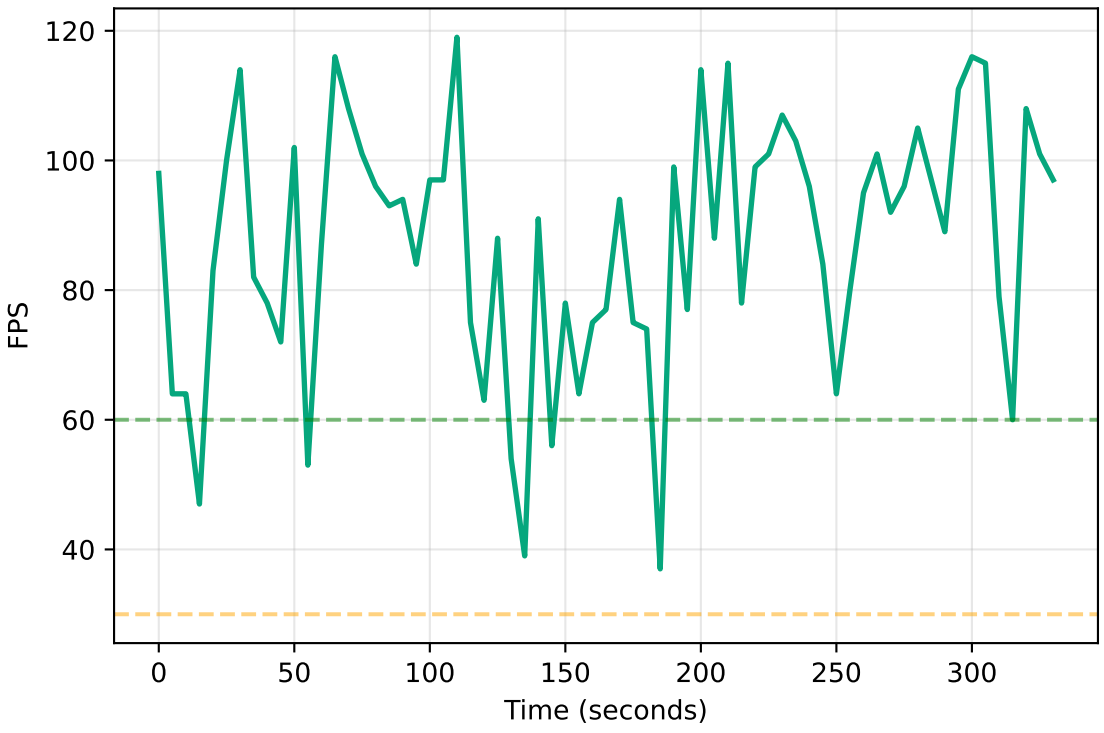
```
MEMORY GROWTH SUMMARY
Starting Memory:  560.0 MB
Ending Memory:    4915.2 MB
Total Growth:     +4355.2 MB
Percent Increase: +777.7%
Test Duration:    724 seconds
```

Phase-by-Phase Performance Comparison

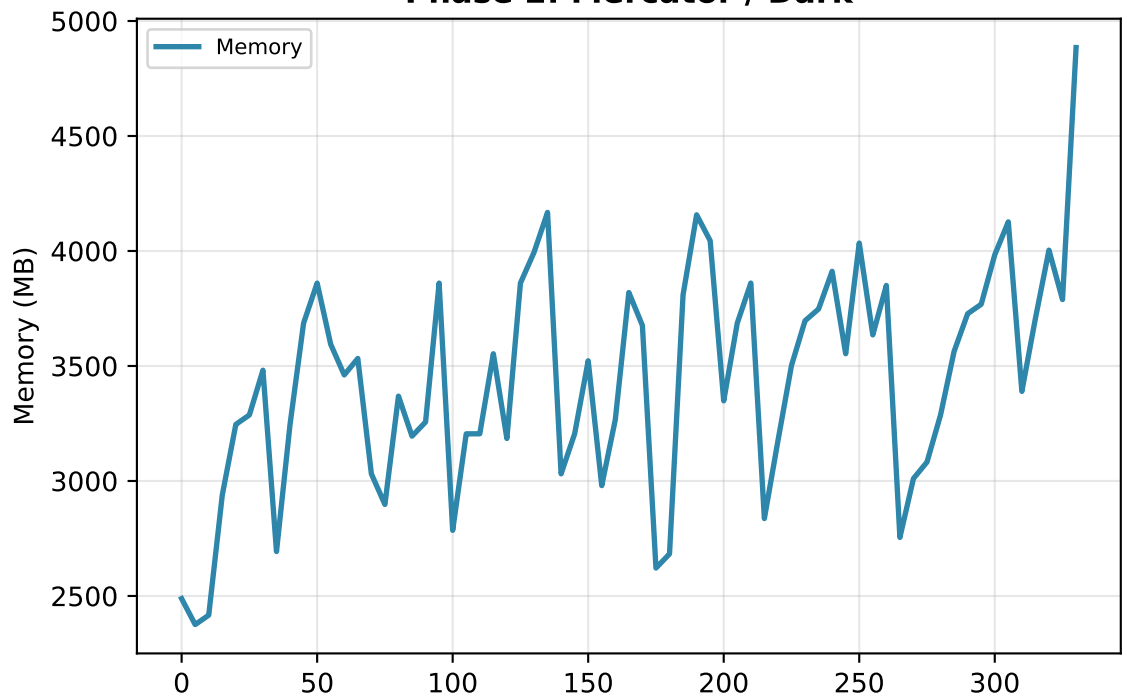
Phase 1: Globe / Satellite

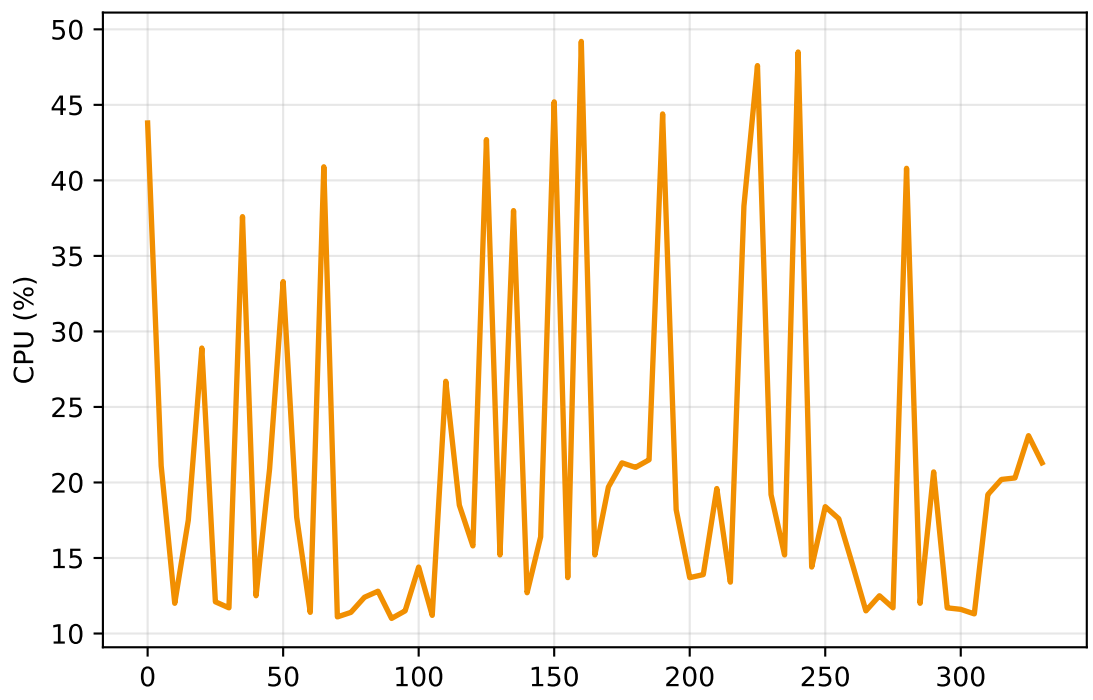


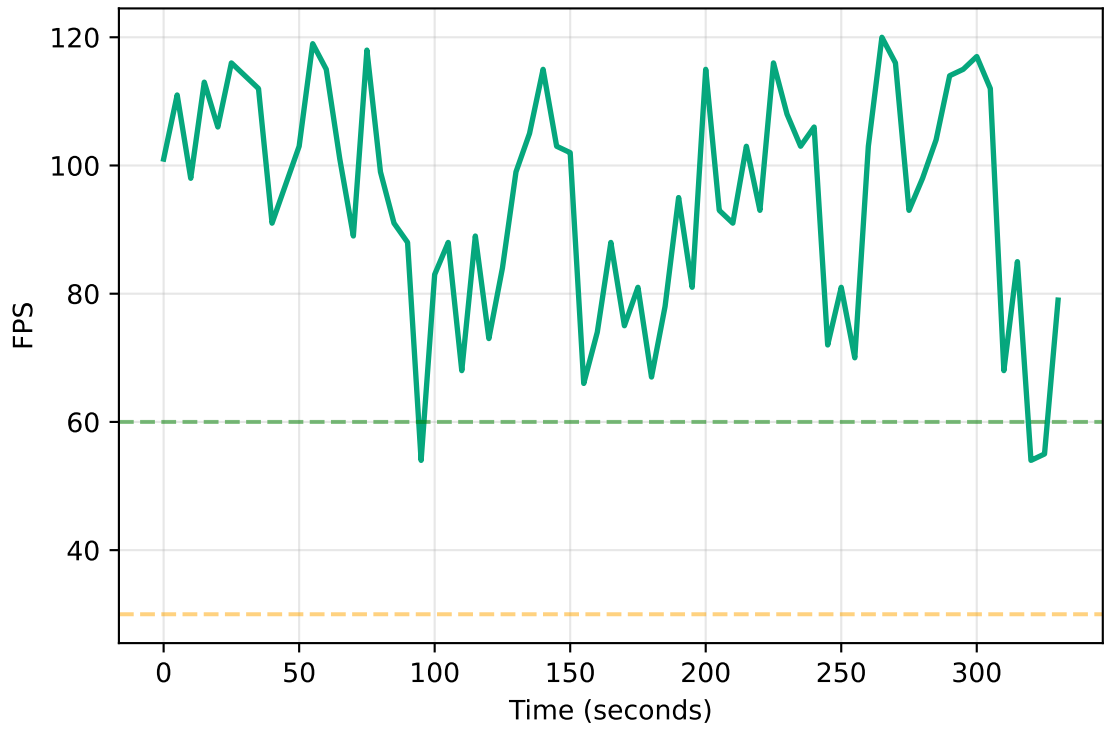




Phase 2: Mercator / Dark

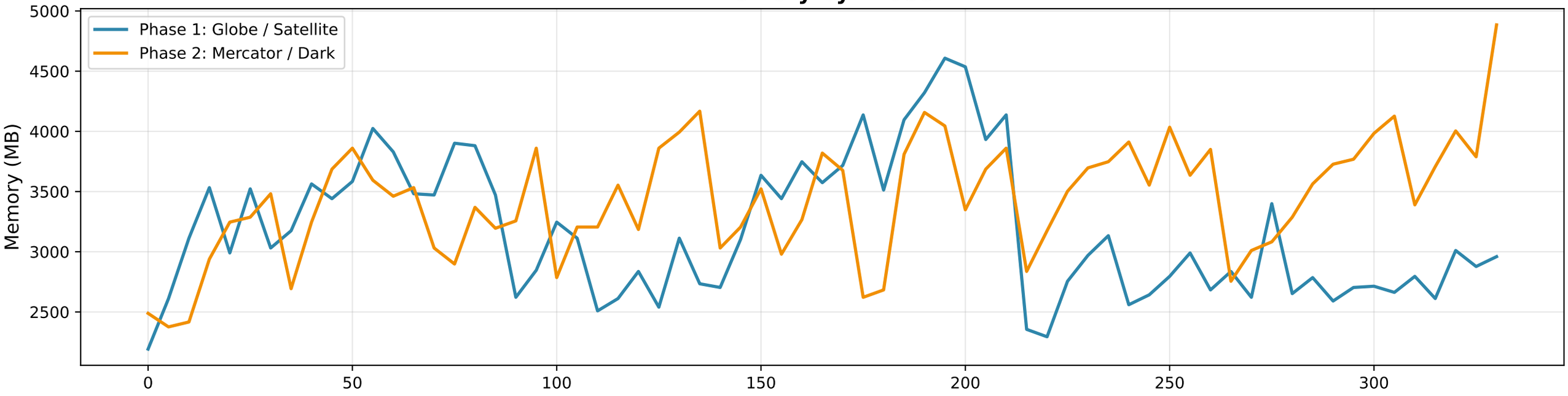




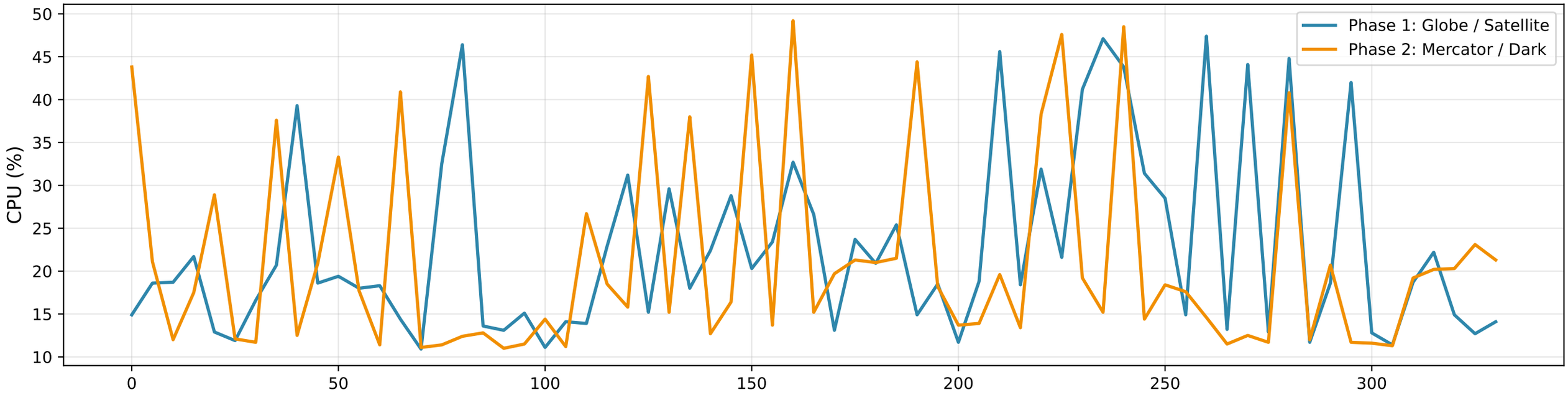


Phase Overlay Comparison

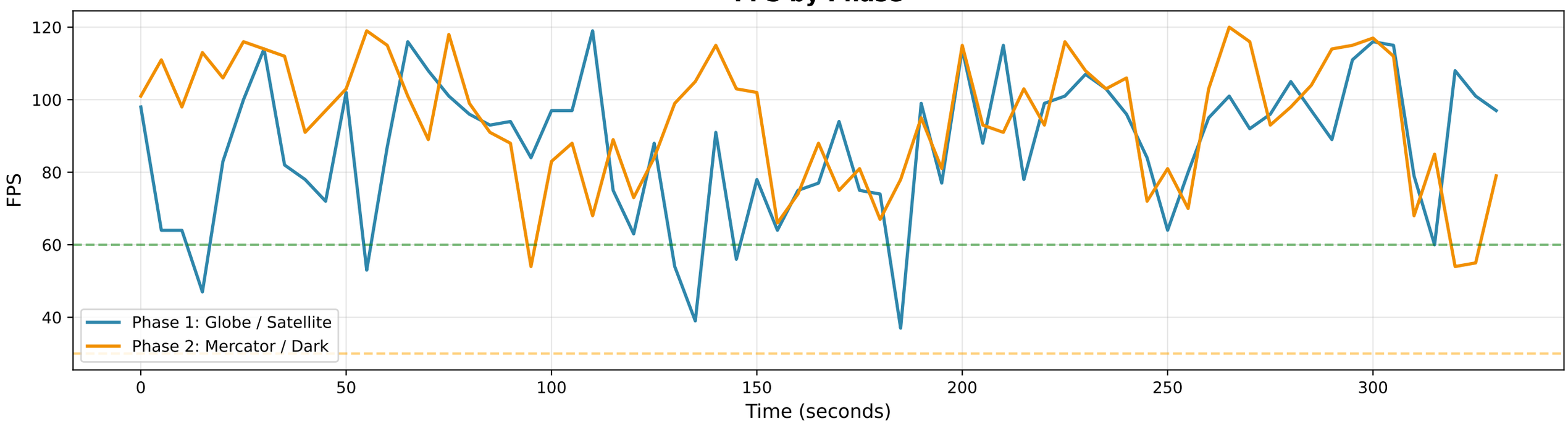
Memory by Phase



CPU by Phase

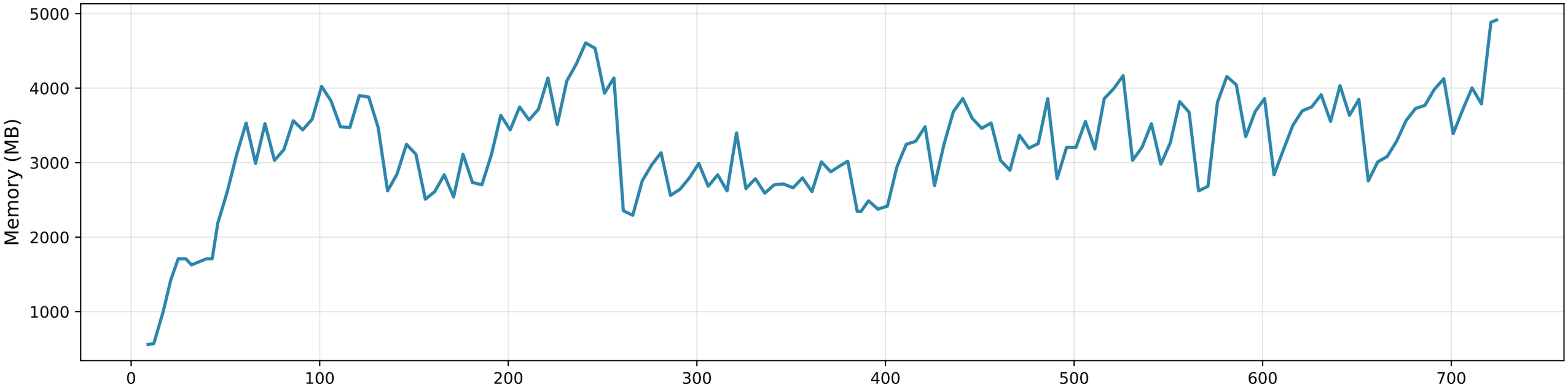


FPS by Phase

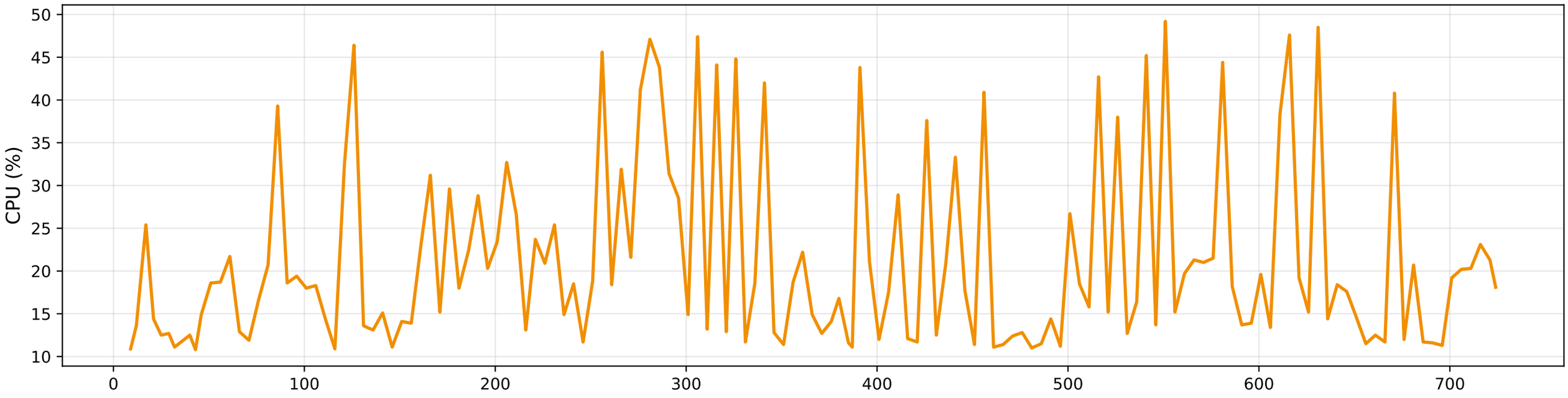


Combined Performance Timeline

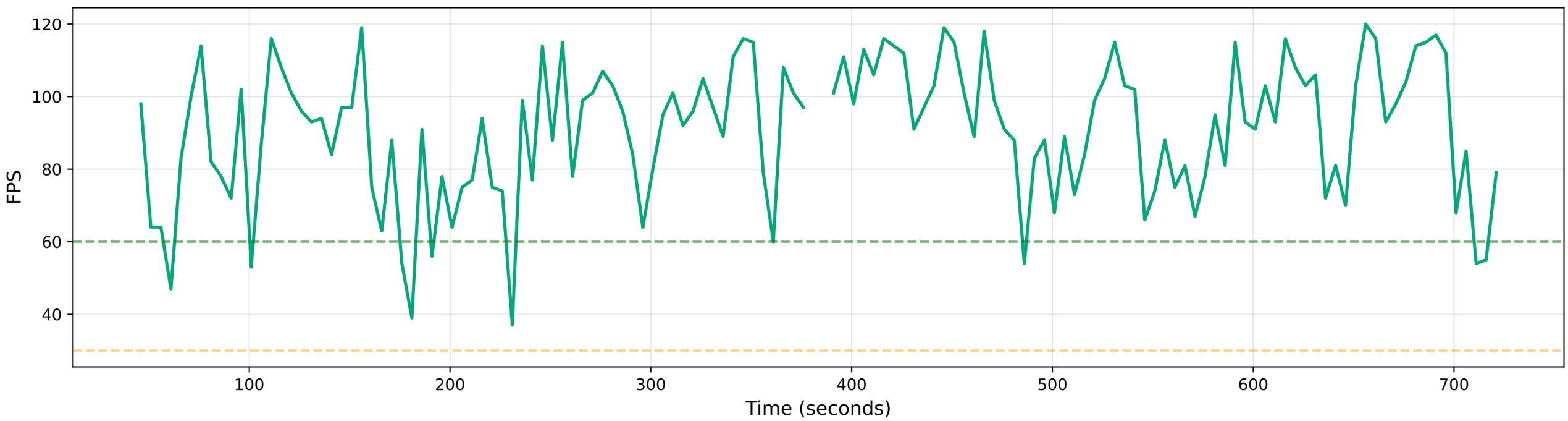
Memory Over Time

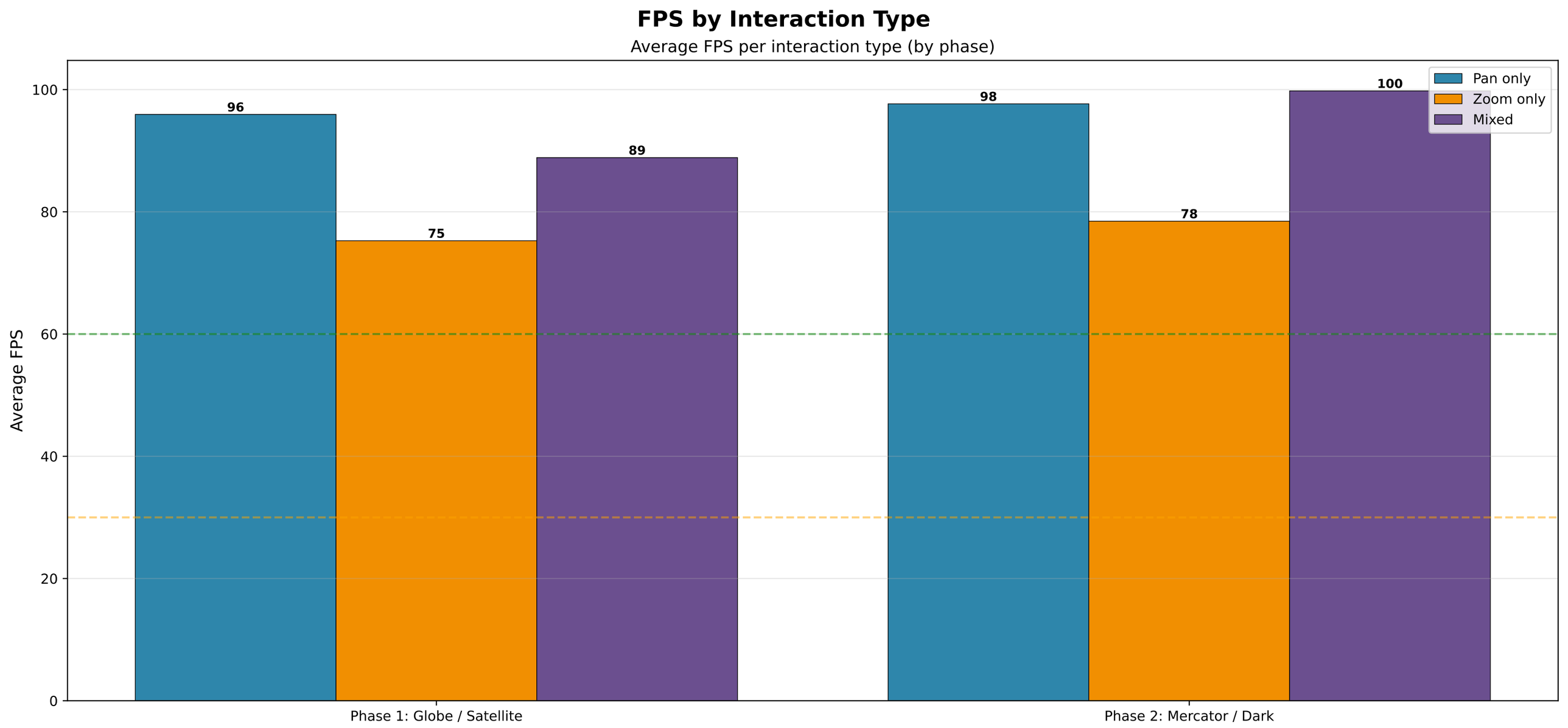


CPU Over Time



FPS Over Time





Phase	Pan only	Zoom only	Mixed
Phase 1: Globe / Satellite	96	75	89
Phase 2: Mercator / Dark	98	78	100

Console Errors by Frequency

No console errors captured during this test